

Data and Analysis Repository

This repository contains the data and analyses scripts for a Randomized Controlled Trial evaluating Indoor Residual Spraying (IRS) or Behavioral Change Communication in combination to Long-Lasting Insecticidal Nets (LLIN) in Burkina Faso (BF) and Côte d'Ivoire (CI).

Results of this RCT have been published in: Moiroux et al. Non-pyrethroid indoor residual spraying or intensive behavior change communication in combination with long-lasting insecticidal nets against malaria disease in West Africa: a pragmatic transnational cluster-randomized controlled trial. 2025, The Lancet Global Health.

Suggested citation: Moiroux Nicolas, Replication data and codes for: Non-pyrethroid indoor residual spraying or intensive behavior change communication in combination with long-lasting insecticidal nets against malaria disease in West Africa: a pragmatic transnational cluster-randomized controlled trial. June 2025. Available at: https://github.com/Nmoiroux/REACT_TLGH

Repository Structure

- README.md # This file
- metadata.Rmd # R Markdown file of metadata and dataset descriptions
- Data/ # Folder containing all .RData databases
- R/ # Folder containing all R scripts used in the analyses

Metadata Overview

All datasets are described in `metadata.Rmd`. The `metadata.Rmd` provides a structured summary of all datasets.

Reproducibility

- All datasets are available as R Objects in `.RData` databases and stored in the `Data/` folder.
 - Each R script is designed to be run independently and includes code to load data from the corresponding `.RData` database. We recommend setting the working directory to the root of this repository and sourcing each script from the `R/` folder.
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R Scripts and Required Datasets

R Script File	Analysis	Required Datasets	.Rdata database
<code>Meta_analysis_nonPYR_IRS_update.R</code>	Non-PYR IRS Meta-analysis	<code>data_meta_IRS</code>	<code>data_meta_IRS.RData</code>
<code>Primary_outcome_Incidence_rate_Analysis.R</code>	Incidence rates (Passive detection)	<code>pas_ana_rate_subgroup</code>	<code>data.RData</code>
<code>Secondary_outcome_Anopheles_EIR_Analysis.R</code>	Entomolgical Incolutation Rate	<code>eir_BF, eir_CI</code>	<code>entomo.RData</code>

R Script File	Analysis	Required Datasets	.Rdata database
Secondary outcome_Anopheles Exophagy_Analysis.R	Exophagy	ER	entomo.RData
Secondary outcome_Anopheles HBR_Analysis.R	Human Biting Rate (Anopheles)	hbr	entomo.RData
Secondary outcome_Anopheles Hours_Analysis.R	Biting times (Anopheles)	Hour_BF, Hour_CI	entomo.RData
Secondary outcome_Anopheles PR_Analysis.R	Parity Rate (Anopheles)	PR_BF, PR_CI	entomo.RData
Secondary outcome_Anopheles Resistances_Analysis.R	kdr-w, kdr-e & ace-1 alleles freq.	KDRW_BF, KDRW_CI, KDRE_BF, ACE1_BF, ACE1_CI	entomo.RData
Secondary outcome_Anopheles SR_Analysis.R	Sporozoite Rate (Anopheles)	SR_BF, SR_CI	entomo.RData
Secondary outcome_Case prevalence_Analysis.R	Case prevalence (Survey)	df_case	data.RData
Secondary outcome_Infection Prevalence_Analysis.R	Infection prevalence (Survey)	df_GE_BF, df_GE_CI	data.RData
Secondary outcome_LLIN use_Analysis.R	LLIN use rate	df_use	data2.RData
Secondary outcome_Mosquito HBR_Analysis.R	Human Biting Rate (all mosquitoes)	nuis	data.RData
Secondary outcome_Parasite density_Analysis.R	Parasite density (Survey)	df_dp_BF, df_dp_CI	data.RData

Notes

- Please cite appropriately if using this data or code.
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